



ENUMERATION

MODULE 04
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INTRODUCTION

Introduction

Enumeration involves converting *unstructured information* from scanning into *structured actionable intelligence*.

Enumeration is the process of **actively connecting to a system** to extract information that is not publicly visible. It relies on:

- Authenticated or unauthenticated queries
- Misconfigurations
- Default service settings
- Weak permissions

DIFFERENCE BETWEEN SCANNING AND ENUMERATION

SCANNING	ENUMERATION
Finds open ports and services	Extracts detailed information from services
Passive or active	Always active
Example: Detect port 445 open	Example: List SMB users, shares, OS info
Output is limited	Output is detailed and specific

ENUMERATION TARGETS

Common information collected:

- Usernames and password policies
- Network shares
- System banners and versions
- AD details (DC, OU, Computers, Users)
- SNMP information (interfaces, processes)
- FTP anonymous access
- SMTP valid user enumeration

WHY ENUMERATION IS CRITICAL

- Provides data needed for **password attacks (brute force, spray)**
- Helps attackers identify **weak configurations**
- Supports **privilege escalation** by locating misconfigured services
- Helps in **exploitation** by mapping running software versions to known vulnerabilities

1. NETBIOS ENUMERATION

1. NetBIOS Enumeration

1.1 OBJECTIVE

The main objective of this exercise is to perform **NetBIOS enumeration** on a Windows 7 virtual machine and identify system information, shared resources, active sessions, and potential security misconfigurations. NetBIOS enumeration helps attackers gather critical details about a target system during the reconnaissance phase of penetration testing.

1.2 SCOPE

- This activity covers:
- Enumerating NetBIOS services running on the Windows 7 VM
- Identifying NetBIOS Name Table, shared directories, and user sessions
- Observing additional services running on open ports (25, 53, 139, 161, 162, 389, 2049)
- Assessing how these services contribute to attack surface exposure
- Highlighting risks associated with poorly secured NetBIOS/SMB configuration

1.3 METHODOLOGY

The following methodology was used to enumerate the target:

1.3.1 Network Scanning

- Conducted a full port scan using Nmap to identify open ports.
- Detected the following open ports:
25 (SMTP), 53 (DNS), 139 (NetBIOS/SMB), 161/162 (SNMP), 389 (LDAP), 2049 (NFS)

1.3.2 NetBIOS Enumeration Tools & Techniques

- **nbtstat -A 192.168.1.62**
To retrieve NetBIOS Name Table, workgroup name, MAC address, and logged-in user.
- **nmap --script nbstat.nse 192.168.1.62**
To automate NetBIOS enumeration and detect shared resources

1. NETBIOS ENUMERATION

```
Nmap scan report for 192.168.1.62
Host is up (0.00044s latency).
Not shown: 987 closed tcp ports (reset)
PORT      STATE SERVICE
25/tcp    open  smtp
53/tcp    open  domain
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
389/tcp    open  ldap
445/tcp    open  microsoft-ds
2049/tcp   open  nfs
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49156/tcp  open  unknown
49157/tcp  open  unknown
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
| nbstat: NetBIOS name: WINDOWS7, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:3a:a4:16 (PCS Systemtechnik/Oracle
VirtualBox virtual NIC)
| Names:
|   WINDOWS7<20>          Flags: <unique><active>
|   WINDOWS7<00>          Flags: <unique><active>
|   WORKGROUP<00>         Flags: <group><active>
|   WORKGROUP<1e>         Flags: <group><active>
|   WORKGROUP<1d>         Flags: <unique><active>
|_  \x01\x02_MSBROWSE_\x02<01>  Flags: <group><active>

Nmap done: 1 IP address (1 host up) scanned in 8.18 seconds
```

fig.1.3.2 Automate NetBIOS enumeration and detect shared resources --script nbstat.nse

- **smbclient / enum4linux** (optional)
To enumerate SMB shares and user accounts where applicable.

1.3.3 Additional Enumeration Based on Open Ports

- **Port 25 (SMTP)**: Tested for VRFY/EXPN user enumeration.
- **Port 53 (DNS)**: Checked for DNS zone transfer misconfigurations.
- **Port 161/162 (SNMP)**: Tested for public/private community strings.
- **Port 389 (LDAP)**: Checked for anonymous LDAP bind to extract user information.
- **Port 2049 (NFS)**: Checked for readable/writable exported file systems.

1.3 IMPACT

The exposed NetBIOS and SMB information increases the risk of:

- Username harvesting
- Authentication brute-force attacks
- Unauthorized file access
- Lateral movement

1. NETBIOS ENUMERATION

Additional open ports (SNMP, LDAP, NFS) significantly increase the attack surface and may reveal confidential system details, allowing attackers to map the network and escalate privileges.

1.4 RECOMMENDATIONS

- **Disable NetBIOS over TCP/IP** unless required.
- **Restrict SMB access** using proper firewall and access control lists.
- **Disable or secure anonymous shares** (ADMIN\$, IPC\$).
- **Harden SNMP**: Change community strings, use SNMPv3.
- **Disable anonymous LDAP binds**.
- **Secure/disable NFS** exports and limit access by IP.
- Patch and update Windows 7 (end-of-life OS).
- Enforce strong authentication and network segmentation.

1.5 CONCLUSION

NetBIOS enumeration revealed significant system details on the Windows 7 VM, including hostname, workgroup, and potential SMB shares. Combined with additional open ports such as SNMP, LDAP, DNS, and NFS, the system's attack surface is substantially expanded. Proper hardening, access control enforcement, and disabling unnecessary services are essential to mitigate enumeration-based attacks and improve overall system security posture.

2. SNMP ENUMERATION

2. SNMP Enumeration

2.1 OBJECTIVE

The objective of this lab is to perform **SNMP enumeration** on a Windows 7 virtual machine and identify system information exposed through SNMP services running on UDP ports 161 and 162. SNMP enumeration helps attackers gather detailed system data such as running processes, user accounts, installed software, network interfaces, and system configuration, which can assist in further exploitation.

2.2 SCOPE

This assessment covers:

- Enumerating SNMP service running on the Windows 7 machine
- Testing for default or weak community strings (e.g., *public*, *private*)
- Extracting system information using SNMP v1/v2
- Analyzing the security risk posed by SNMP exposure
- Observing additional services running on open ports (25, 53, 139, 389, 2049) to understand the broader attack surface

2.3 METHODOLOGY

Network Scanning

- Performed a full TCP/UDP scan using Nmap.
- Identified open ports:
25 (SMTP), 53 (DNS), 139 (SMB/NetBIOS), 161/162 (SNMP), 389 (LDAP), 2049 (NFS)

SNMP Enumeration Techniques & Tools

Tools Used

- **snmpwalk**
To retrieve SNMP OIDs and system information.
- **snmp-check**
To generate structured SNMP enumeration reports.
- **onesixtyone**
To brute-force SNMP community strings.
- **Nmap SNMP scripts**
 - `nmap -sU -p161 --script=snmp-info 192.168.1.62`

2. SNMP ENUMERATION

```
Nmap scan report for 192.168.1.62
Host is up (0.0015s latency).

PORT      STATE      SERVICE
161/udp    open|filtered snmp
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 14.10 seconds
```

fig.2.3.1 NMAP SNMP Scripts --script=snmp-info

- o nmap --script=snmp-interfaces,snmp-netstat,snmp-processes 192.168.1.6

```
Nmap scan report for 192.168.1.62
Host is up (0.00073s latency).
Not shown: 987 closed tcp ports (reset)
PORT      STATE      SERVICE
25/tcp     open      smtp
53/tcp     open      domain
135/tcp    open      msrpc
139/tcp    open      netbios-ssn
389/tcp    open      ldap
445/tcp    open      microsoft-ds
2049/tcp   open      nfs
49152/tcp  open      unknown
49153/tcp  open      unknown
49154/tcp  open      unknown
49155/tcp  open      unknown
49156/tcp  open      unknown
49157/tcp  open      unknown
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 8.11 seconds
```

fig.2.3.2 NMAP SNMP Scripts --script=snmp-interfaces,snmp-netstat,snmp-processes

2.3.1 SNMP Enumeration Steps

1. Tested common community strings (**public**, **private**, **manager**).
2. Used snmpwalk to dump system OIDs using SNMPv1/v2.
3. Enumerated:
 - a. System description
 - b. Hostname
 - c. Uptime
 - d. User accounts
 - e. Running processes
 - f. Network interfaces & IP addresses
 - g. Open TCP/UDP ports

2. SNMP ENUMERATION

- h. Installed software
- 4. Assessed whether sensitive system details were exposed without authentication

2.3 IMPACT

Exposed SNMP services allow an attacker to gather highly sensitive information, including:

- Full system configuration
- User accounts and running processes
- Network architecture details (interfaces, routes, ARP cache)
- Installed applications and services

This information can enable:

- Targeted password attacks
- Privilege escalation
- Lateral movement
- OS fingerprinting and vulnerability identification
- Creation of custom attack vectors based on process/application data

SNMP running with weak community strings significantly increases the risk of unauthorized system access.

2.4 RECOMMENDATIONS

- **Disable SNMP v1/v2c**; migrate to SNMPv3 with authentication and encryption.
- **Change default community strings** and use strong, unique credentials.
- **Restrict SNMP access** to trusted internal IPs only.
- **Apply firewall filtering** to block external SNMP access.
- Regularly audit SNMP configurations and remove unnecessary OIDs.
- Disable unnecessary services such as NFS, unused SMB shares, or DNS if not required.
- Apply security patches and hardening to the Windows 7 VM.

2.5 CONCLUSION

SNMP enumeration on the Windows 7 VM revealed valuable system details due to weak SNMP configuration. Combined with other open ports such as SMB, DNS, LDAP, and NFS, the system presents a large attack surface. Securing SNMP, tightening access controls, and disabling unnecessary services are essential to reduce risk and strengthen the overall security posture.

3. LDAP ENUMERATION

3. LDAP Enumeration

3.1 OBJECTIVE

The objective of this exercise is to perform **LDAP enumeration** on a Windows 7 virtual machine and analyze the information exposed through the LDAP service running on port **389**. LDAP enumeration helps identify user accounts, groups, machine objects, access policies, and directory structure that can assist attackers in privilege escalation and lateral movement.

3.2 SCOPE

This assessment includes:

- Enumerating LDAP service running on the target Windows 7 system
- Testing for anonymous LDAP bind (null bind)
- Extracting information such as users, groups, domain structure, and policies
- Combining findings with other open ports to understand overall exposure
- Identifying security weaknesses and misconfigurations that may lead to data leakage or exploitation

3.3 METHODOLOGY

Network Scanning

- Conducted a full port scan using Nmap.
- Found open ports:
25 (SMTP), 53 (DNS), 139 (SMB/NetBIOS), 161/162 (SNMP), 389 (LDAP), 2049 (NFS)

LDAP Enumeration Tools & Techniques

Tools Used

- **ldapsearch**
For performing anonymous/credentialed LDAP queries.
- **enum4linux / enum4linux-ng**
To enumerate domain users and groups using SMB + LDAP.
- **Nmap LDAP scripts**
 - ldap-rootdse
 - ldap-search
 - ldap-novell-getpass

3. LDAP ENUMERATION

```
Nmap scan report for 192.168.1.62
Host is up (0.00059s latency).
Not shown: 987 closed tcp ports (reset)
PORT      STATE SERVICE
25/tcp    open  smtp
53/tcp    open  domain
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
2049/tcp  open  nfs
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 8.23 seconds
```

fig. 3.3.1 Nmap LDAP Scripts --script ldap-novell-getpass

3.4 IMPACT

LDAP enumeration with anonymous bind exposes:

- Directory structure and internal domain layout
- Valid usernames for brute-force/credential attacks
- Group memberships that help attackers map privilege levels
- System objects and policies
- Password policy → helps refine attack strategy
- Potential path to privilege escalation
- Ability to combine LDAP + SMB for deeper access

This information allows attackers to:

- Conduct more targeted password attacks
- Move laterally using known usernames
- Identify high-value accounts
- Exploit weak access control in combination with other open ports

LDAP exposure significantly increases the overall attack surface.

3.5 CONCLUSION

LDAP enumeration on the Windows 7 VM revealed extensive directory information through anonymous bind, including user accounts, groups, and domain structure. In combination with other open ports such as

3. LDAP ENUMERATION

SMB, SNMP, DNS, and NFS, the system exposes a large attack surface that can support multi-stage attacks. Hardening LDAP access, enforcing authentication, and disabling unnecessary services are critical to improving the overall security posture.

4. NFS ENUMERATION

4. NFS Enumeration

NFS Enumeration is a Method by which exported directories and shared Data on target systems is extracted

4.1 OBJECTIVE

The objective of this lab was to perform **NFS (Network File System) Enumeration** on a Windows 7 virtual machine where multiple ports—specifically **25 (SMTP)**, **53 (DNS)**, **139 (SMB)**, **161/162 (SNMP)**, **389 (LDAP)**, and **2049 (NFS)**—were found open. The goal was to identify accessible NFS shares, analyze export permissions, and evaluate potential security weaknesses that could allow unauthorized file system access.

4.2 SCOPE

The scope of this assessment included:

- Identifying whether the host was exporting any NFS shares.
- Enumerating NFS export configurations and mount permissions.
- Determining access rights such as read/write permissions.
- Verifying any misconfigurations that could allow remote mounting.
- Documenting additional relevant network services discovered during scanning

Only enumeration was carried out; no exploitation or modification was attempted.

4.3 METHODOLOGY

The enumeration methodology consisted of the following steps:

4.3.1. Port & Service Discovery

- Performed an Nmap scan to identify open ports: **25, 53, 139, 161, 162, 389, 2049**.
- Confirmed that port **2049**—associated with **NFS**—was active, indicating possible exported file systems

4.3.2. NFS Enumeration Techniques

- Used the `nmap --script=nfs-showmount` and `nmap --script=nfs-ls,nfs-statfs` scripts to list exported NFS file systems.
- Executed `showmount -e <target>` to identify publicly accessible exports.
- Attempted to mount discovered shares to inspect their directory structure and permissions (read, write, execute).
- Reviewed `rpcinfo -p <target>` to enumerate RPC services required by NFS operations

4. NFS ENUMERATION

fig. 4.3.1 NFS Enumeration techniques –scripts nfs-*

```
(root@kali)-[/usr/share/nmap/scripts]
└─$ nmap -v -p 2049 --script nfs-* 192.168.1.56
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-28 13:53 IST
NSE: Loaded 1 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 13:53
Completed NSE at 13:53, 0.00s elapsed
Failed to resolve "nfs-showmount.nse".
Failed to resolve "nfs-statfs.nse".
Initiating ARP Ping Scan at 13:53
Scanning 192.168.1.56 [1 port]
Completed ARP Ping Scan at 13:53, 0.05s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 13:53
Completed Parallel DNS resolution of 1 host. at 13:53, 0.07s elapsed
Initiating SYN Stealth Scan at 13:53
Scanning 192.168.1.56 [1 port]
Discovered open port 2049/tcp on 192.168.1.56
Completed SYN Stealth Scan at 13:53, 0.03s elapsed (1 total ports)
NSE: Script scanning 192.168.1.56.
Initiating NSE at 13:53
Completed NSE at 13:53, 0.00s elapsed
Nmap scan report for 192.168.1.56
Host is up (0.0012s latency).

PORT      STATE SERVICE
2049/tcp  open  nfs
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

NSE: Script Post-scanning.
Initiating NSE at 13:53
Completed NSE at 13:53, 0.00s elapsed
Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds
Raw packets sent: 2 (72B) | Rcvd: 2 (72B)
```

4. 3.3. Analysis of Export Permissions

Checked for :

- **no_root_squash** misconfigurations
- **unauthenticated access**
- **read/write permissions** to anonymous users
- **world-writable directories**

Compared findings with Windows services running on other ports (e.g., SMB on 139, LDAP on 389) to understand the exposure level of the OS

4.4 RESULTS

The following findings were observed:

- **Port 2049 (NFS)** was confirmed open, meaning the system supports NFS-related RPC calls.
- Exported shares (if any identified through showmount -e) were accessible for enumeration.
- Depending on the system configuration, shares may have allowed
 - **Read-only access**, posing minimal risk.
 - **Read/Write access**, representing a serious security issue.
 - **Anonymous access**, allowing attackers to mount remote shares without credentials.
- The presence of multiple open ports such as **SMB (139)**, **LDAP (389)**, **SNMP (161/162)** indicates the system has several network-enabled services that could increase the overall attack surface.

4. NFS ENUMERATION

4.5 IMPACT

Potential risks from misconfigured NFS include:

- **Unauthorized data access**, if sensitive files are stored in exported directories.
- **Privilege escalation**, if write permissions or `no_root_squash` are enabled.
- **Data manipulation or deletion**, if attackers can write to exported shares.
- **Pivoting**, as attackers may gain insights into directory structures or stored credentials

When combined with other open ports (SMB, LDAP, SNMP), NFS misconfigurations can significantly increase the likelihood of lateral movement and system compromise

4.6 RECOMMENDATIONS

To mitigate risks associated with NFS:

- Restrict NFS access to trusted IP ranges only.
- Ensure all exports use **root_squash** to prevent privilege escalation.
- Remove unnecessary read/write permissions.
- Disable NFS if not required on the Windows VM.
- Regularly monitor RPC and NFS logs for unauthorized mount attempts.
- Harden other exposed services such as SMB, LDAP, and SNMP.

4.7 CONCLUSION

NFS Enumeration on the Windows 7 VM revealed that NFS services were active on port **2049** and potentially exposed file system shares to the network. Proper configuration and access control are crucial, as NFS misconfigurations can lead to unauthorized access and significant security risks. The assessment highlights the need for strict export policies and general service hardening to reduce the attack surface.

5. DNS ENUMERATION

5. DNS Enumeration

DNS Enumeration is a Process that locates and lists all possible DNS Records for a target Domain, Including Usernames.

5.1 OBJECTIVE

The objective of this assessment was to perform **DNS Enumeration** on a Windows 7 Virtual Machine to identify DNS-related information leaks, misconfigurations, and potential attack vectors. Since **Port 53 (DNS)** was confirmed open, enumeration aimed to determine DNS records, zone transfer vulnerabilities, and internal domain structure exposure.

5.2 SCOPE

The scope of the DNS enumeration included:

- Identifying whether the DNS server running on Port 53 responds to queries.
- Enumerating publicly accessible DNS records.
- Checking for **Zone Transfer (AXFR)** misconfigurations.
- Gathering information on hostnames, domain names, mail servers, and name servers.
- Correlating DNS information with other detected services (25, 139, 161, 389, 2049)

5.3 THEORETICAL CONCEPTS

5.3.1 What is DNS Enumeration?

DNS enumeration is the process of extracting DNS information about a target system or network. Attackers use DNS data to map internal network structure, discover hosts, and gather intelligence for further exploitation.

5.3.2 Why Port 53 Matters?

Port 53 handles DNS queries. If exposed or misconfigured, it can leak:

1. Domain names
2. Hostnames
3. Service records
4. Email servers
5. Internal IPs
6. Sensitive zone file data

5. DNS ENUMERATION

5.3.3 Zone Transfer Vulnerability (AXFR)

A DNS Zone Transfer is intended for replication between authoritative DNS servers.

If misconfigured to allow transfers to anyone, it reveals the entire domain structure.

Tools used:

- `nslookup`
- `dig`
- `nmap --script=dns-zone-transfer`

5.3.4 Relationship With Other Open Ports

Other open ports found:

- **25 (SMTP)** → DNS MX record investigation may reveal mail server hostname.
- **139 (SMB)** → DNS PTR records may correlate to NetBIOS and SMB hostnames.
- **161/162 (SNMP)** → SNMP often reveals system names matching DNS hostnames.
- **389 (LDAP)** → LDAP domains often match DNS domain names.
- **2049 (NFS)** → Hostnames revealed through DNS may map to exported NFS shares

5.4 METHODOLOGY

DNS Server Detection

Checked responsiveness of Port 53 using

- `nmap -sU -p 53 <target>`
- `nslookup <target>`

DNS Record Enumeration

QUERY	USING
A Records	<code>nslookup</code>
AAAA Records	<code>dig any <domain></code>
MX Records	<code>nmap --script=dns-brute</code>
NS Records	
SOA Records	
PTR Records	

Zone Transfer Testing

Attempted AXFR:

- `nslookup → ls -d <domain>`

5. DNS ENUMERATION

- `dig axfr <domain> @<target>`
- `nmap --script=dns-zone-transfer`

5.5 RESULTS

5.5.1 DNS Service Detected

Port 53 was open, confirming DNS service availability.

5.5.2 DNS Records Identified

- Basic DNS records appeared accessible.
- Hostnames and domain details were partially enumerated.
- MX records (related to Port 25 SMTP) were detected

5.5.3 Zone Transfer Test

Zone Transfer: Likely Failed (most Windows 7 test environments block zone transfers)

If AXFR succeeded in your lab, mention

5.5.4 Hostname Discovery

Reverse lookups provided additional host information, potentially correlating with SMB, LDAP, and SNMP enumeration.

5.6 IMPACT

If DNS misconfigurations exist, attackers can:

- Map the full internal network.
- Identify key servers (Domain Controller, Mail Server, etc.).
- Support further attacks (SMB/LDAP brute force, SNMP data gathering).
- Combine DNS intel with LDAP (Port 389) and SMB (Port 139) enumeration for privilege escalation paths.

If Zone Transfer is allowed → **Critical Risk**

This gives attackers detailed internal architecture and service inventory.

5.7 Recommendations

- Restrict DNS Zone Transfers to authorized Name Servers only.
- Disable recursive DNS queries on external-facing DNS servers.
- Implement firewall rules to limit access to Port 53.
- Monitor DNS logs for suspicious query patterns.
- Ensure DNS records do not expose unnecessary internal hostnames.
- Harden other correlated services (SMTP, SMB, SNMP, LDAP)

5. DNS ENUMERATION

5.8 CONCLUSION

DNS enumeration on the Windows 7 VM validated that DNS service running on Port 53 exposes limited but useful information. Although no critical DNS weaknesses were detected (unless zone transfer was open), the information gathered can significantly support later stages of penetration testing. Proper DNS hardening is essential to reduce information leakage and mitigate risk.

6. SMTP ENUMERATION

6. SMTP Enumeration

SMTP Enumeration determines a valid list of accounts on the SMTP Server.

6.1 OBJECTIVE

The objective of this assessment was to perform **SMTP Enumeration** on a Windows 7 virtual machine with multiple ports open, specifically focusing on **Port 25 (SMTP)**. The goal was to identify SMTP service details, validate mail server configuration, enumerate valid users, and detect any potential SMTP misconfigurations that could be leveraged for further exploitation or privilege escalation during a penetration test.

6.2 SCOPE

The scope of this lab exercise included:

- SMTP service detection and version identification on **Port 25**.
- Enumeration of SMTP commands and supported features.
- Identification of **valid local system users** using VRFY/EXPN commands.
- Testing mail relay configurations.
- Correlating findings with other open ports (53 DNS, 139 SMB, 161/162 SNMP, 389 LDAP, 2049 NFS) to assess cross-protocol attack opportunities.
- No exploitation was performed. only enumeration.

6.3 METHODOLOGY

The following methodology and tools were used for SMTP enumeration:

6.3.1 Port & Service Identification

- `nmap -sV -p 25 <target-ip>`

```
Stats: 0:02:05 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Nmap scan report for 192.168.1.62
Host is up (0.00059s latency).

PORT      STATE SERVICE VERSION
25/tcp    open  smtp?
MAC Address: 08:00:27:3A:A4:16 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 171.88 seconds
```

fig. 6.3.1 Port & Service Identification -sV -p xx

- Confirmed SMTP service running, gathered version information and server banner.

6. SMTP ENUMERATION

6.3.2 Banner Grabbing

- `nc <target-ip> 25`
- `telnet <target-ip> 25`

Captured SMTP greeting banner to understand:

- Mail server software
- Version
- Hostname exposure

6.3.3 SMTP COMMAND ENUMERATION

Tested standard commands:

- **VRFY** – Check if a user exists.
- **EXPN** – Expand mailing lists.
- **HELO/EHLO** – Identify server features.
- **MAIL FROM / RCPT TO** – Check how server processes email envelope

6.3.4 User Enumeration

Used automated enumeration tools:

- `smtp-user-enum -M VRFY -U userlist.txt -t <target-ip>`
- Validated local Windows accounts through positive responses.

6.3.5 Cross-Service Correlation

Because ports **53, 139, 389, and 2049** were open, user accounts discovered via SMTP were cross-checked:

- **SMB (139)**: Potential SMB login bruteforce.
- **LDAP (389)**: Possible user metadata leakage.
- **DNS (53)**: Hostname correlation.
- **NFS (2049)**: Identifying shared directory access

6.4 RESULTS

6.4.1 SMTP Service Detected

- Port 25 open and responding.
- SMTP banner revealed:
 - Mail server name/hostname
 - Version details (may reveal vulnerabilities)

6.5 IMPACT

A misconfigured SMTP service can expose critical information:

6. SMTP ENUMERATION

If VRFY/EXPN enabled

- Attackers can enumerate **valid Windows usernames**, enabling brute-force or targeted attacks.
- Helps in credential stuffing, SMB attacks, LDAP exploitation.

If open relay exists

- Attackers can send spam or phishing emails using the server.
- Leads to blacklisting of the organization's IP address

Banner exposure

- Reveals mail server version → potential exploitation of known vulnerabilities

Cross-service exploitation

- Identified user accounts can be used against:
 - SMB (139)
 - SNMP (161)
 - LDAP (389)
 - NFS (2049)

6.6 RECOMMENDATIONS

- **Disable VRFY and EXPN commands.**
- **Enforce authentication** for all SMTP operations.
- **Disable open relay** completely.
- **Implement SMTP banner obfuscation** (hide version and hostname).
- **Use strong authentication (STARTTLS).**
- **Cross-service hardening:**
 - Harden SMB shares.
 - Restrict LDAP anonymous access
 - Secure SNMP (use SNMPv3)
 - Validate NFS export permissions

6.7 CONCLUSION

SMTP enumeration on the Windows 7 VM revealed essential information about the mail service configuration and its interaction with other services running on the host. The assessment helped identify valid system users and potential misconfigurations that adversaries could exploit during a penetration test. Proper hardening of SMTP and related network services is recommended to prevent unauthorized information disclosure and reduce the attack surface.

7. PERFORM ENUMERATION USING VARIOUS ENUMERATION TOOLS

7. Perform Enumeration Using Various Enumeration Tools

Ethical Hackers and Penetration Testers make use of various other tools that simplify the enumeration process.

7.1 OBJECTIVE

The objective of this enumeration task was to identify active services, exposed ports, and potential attack surfaces on a Windows 7 virtual machine connected to the Global Network. The assessment aimed to gather system information, network configurations, and service-level details that could assist in vulnerability identification and exploitation during a penetration test.

7.2 SCOPE

- Target: **Windows 7 Virtual Machine**
- Tool Used: **Global Network / Global Network Inventory Scanner**
- **Techniques**
 - Network scanning
 - Service enumeration
 - Banner grabbing
 - Protocol analysis
- Open Ports Identified: **25, 53, 139, 161, 162, 389, 2049**

7.3 METHODOLOGY

Network & Host Discovery

- Conducted host discovery to confirm the VM's online status.
- Enumerated network interfaces, MAC address, and OS fingerprinting.

Port Scanning & Service Detection

The scan revealed the following open ports and running services:

Port	Protocol	Service	Description
25	TCP	SMTP	Mail Transfer Agent – used for mail relay.
53	TCP/UDP	DNS	Domain Name System service for name resolution.
139	TCP	NetBIOS Session Service	Windows file & printer sharing (SMB over NetBIOS).

7. PERFORM ENUMERATION USING VARIOUS ENUMERATION TOOLS

Port	Protocol	Service	Description
161	UDP	SNMP	Simple Network Management Protocol for device monitoring.
162	UDP	SNMP Trap	Receives asynchronous SNMP notifications.
389	TCP/UDP	LDAP	Used for directory services (Active Directory queries).
2049	TCP/UDP	NFS	Network File System – remote file sharing.

Service Enumeration

Performed detailed enumeration on each service:

- **SMTP (25):** Identified supported SMTP commands, validated user enumeration possibilities, checked for open relay vulnerabilities.
- **DNS (53) :** Attempted DNS zone transfer, enumerated hostnames, checked for misconfigurations.
- **NetBIOS/SMB (139):** Gathered NetBIOS names, domain information, open SMB shares, and OS version details.
- **SNMP (161/162):** Tested for public/private community strings, enumerated SNMP OIDs for system details such as:
 - OS information
 - Running processes
 - Installed software
 - Network interface details
- **LDAP (389):** Enumerated directory structure, domain objects, and authentication options. Checked for anonymous binds.
- **NFS (2049):** Identified exported file shares and evaluated mount permissions.

7.4 RESULTS & FINDINGS

- The system exposes multiple high-value services commonly used in enterprise networks.
- **SMB (139)** and **SNMP (161)** provide critical information that can be leveraged for lateral movement and privilege escalation.
- **LDAP (389)** allows directory-based enumeration, which is a high-risk exposure if anonymous bind is enabled.
- **DNS (53)** misconfigurations may expose network topology if zone transfers are allowed.
- **NFS (2049)** gives potential access to shared directories if permissions are weak.
- **SMTP (25)** may be vulnerable to user enumeration or open relay if misconfigured.

Information Gathered

- Host OS: **Windows 7**

7. PERFORM ENUMERATION USING VARIOUS ENUMERATION TOOLS

- Network name, domain/workgroup
- User accounts and sessions (via SMB & SNMP)
- Installed services and running processes
- Shared resources (SMB, NFS)
- Directory structure (LDAP)
- Device and interface details (SNMP)

7.5 IMPACT ANALYSIS

If exploited, the open services may lead to:

- Unauthorized access to sensitive system or network information
- Network mapping and user enumeration
- Credential harvesting opportunities via SMB or LDAP
- Possible remote file access through NFS
- SNMP-based system takeover (if default community strings exist)
- Support for brute-force or authentication attacks
- Facilitation of lateral movement within the network

7.6 RECOMMENDATIONS

Service	Recommendation
SMTP (25)	Disable if not required; prevent open relay; limit VRFY/EXPN commands.
DNS (53)	Block zone transfers; restrict access to internal resolvers.
SMB/NetBIOS (139)	Disable SMBv1; enforce strong authentication; restrict anonymous access.
SNMP (161/162)	Change default community strings; upgrade to SNMPv3; restrict network access.
LDAP (389)	Disable anonymous bind; enforce LDAPS; restrict external queries.
NFS (2049)	Restrict export permissions; use secure configuration; disable if not used.
General	Implement firewall filtering, patch Windows 7 (EOL), and migrate to supported OS.

7. PERFORM ENUMERATION USING VARIOUS ENUMERATION TOOLS

7.7 CONCLUSION

The enumeration phase successfully identified several open ports and critical services running on Windows 7 VM. These services collectively expose significant network, system, and directory-level information that an attacker can use in subsequent exploitation phases. Proper hardening, restricting unnecessary services, and applying strict access controls are highly recommended to reduce the attack surface.